EC321 Monetary Economics
BS Economics, Semester 5th
Assignment 1
Money Demand with Real-World Data

Instructions:

- **Data window:** Use annual data from 1999 to 2023 (25 years) for your assigned country**.**
- **Data Source:** World Development Indicators (WDI). Take data of following four series for your assignment:
    - Broad money (current LCU)
    - GDP (current LCU)
    - Consumer price index (2010 = 100)
    - Lending interest rate (%) *(use Deposit interest rate (%) only if lending rate is unavailable)*
    - Keep units as downloaded; do not convert currencies.
- **Missing Data:** For missing data use the following set of rules
    - **i.** Don't fill anything at first. If a year is missing for any series (M, NGDP, CPI, i**),** drop that year**.**
    - **ii.** Use the longest continuous block of years where all four series are present**.** Aim for 1999–2023; if gaps exist, use whatever continuous span you can find (must be ≥15 years).
    - **iii.** If you can't get 15 continuous years**,** switch to the deposit rate (if lending rate is the only problem) and try again. If still <15, then fill year with the average value of the variable**.**
- **Submission deadline:**
    - Email soft copy of your assignment (not more than 5 pages) and the dataset (MS Excel) to **faisalazeem@qau.edu.pk** by October 12, 2025 (Sunday), 11:59 pm.

Question 1: Testing Quantity Theory of Money Demand

Let $\boldsymbol{MV} = \boldsymbol{PY}$. Use data of the country assigned to you to evaluate following implications of the quantity theory:

i. Velocity level and stability

- Compute $V_t = \frac{NGDP_t}{M_t}$. Where $NGDP_t$ is nominal GDP. Calculate the growth rate[1] of V_t and plot it.
- Do predictions of quantity theory of money demand seem to be true for your data? Why or why not?

[1] Growth rate of $V = (V_t - V_{t-1})/V_{t-1}$

ii. Comovement of money and nominal GDP growth

- Compute growth rate of Money[2] (M) and growth rate of Nominal GDP[3]. Plot both series in the same graph with "time" on x-axis.
- Do predictions of quantity theory of money demand seem to be true for your data? Why or why not?

iii. Money growth and inflation

- Construct inflation (π_t) series by taking growth rate of price level[4] for each period. For price level use data of CPI.
- Construct scatter plot of growth rate of money and inflation. Add trend line to it.
- Do predictions of quantity theory of money demand seem to be true for your data? Why or why not?

iv. Give an overall view on the empirical success of quantity theory of money demand for your data.

Question 2 Testing Keynes' Liquidity Preference Theory

According to Keynesian Liquidity Preference Theory:

$$\boldsymbol{\frac{M}{P} = f(y, i)}$$

i. Construct following variables for each year:

- Real money balance= $\frac{M}{P}$. Where $P = CPI/100$
- $Y\ (real\ GDP) = NGDP/P$
- i = lending (or deposit) rate. Keep it in percentage form.

ii. Construct scatter plot of $\frac{M}{P}$ and Y. Is it supporting liquidity preference theory?

iii. Construct scatter plot of $\frac{M}{P}$ and i. Is it supporting liquidity preference theory?

iv. Estimate following model using OLS regression and report results:

$$\ln\left(\frac{M}{P}\right)_t = \alpha + \beta lnY_t + \delta i_t + u_t$$

v. Do your regression results support Keynesian Liquidity Preference Theory or not? Why or why not?

[2] Growth rate of $M = (M_t - M_{t-1})/M_{t-1}$

[3] Growth rate of $NGDP = (NGDP_t - NGDP_{t-1})/NGDP_{t-1}$

[4] $\pi_t == (CPI_t - CPI_{t-1})/CPI_{t-1}$